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RESEARCH INTERESTS

- Soft robotics, muscular-skeletal system
- Origami robots and design, modular origami actuators
- Soft valves, compact soft valve design

EDUCATION

Department of Mechanical Engineering, National University of Singapore Ph.D. in Mechanical Engineering Industrial Ring-fenced Scholarship	2023.8 – Present Singapore
Department of Mechanical Engineering, National University of Singapore M.Sc. in Mechanical Engineering	2022.8 – 2023.6 Singapore
School of Mechanical and Aerospace Engineering, Jilin University B.Eng. in Mechanical Engineering	2018.9 – 2022.6 Changchun, Jilin, China

AWARDS

2018 – 2022: First Class, Second Class and Third Class Academic Scholarship, Faw Jiefang Company Scholarship

PROJECTS

LEGO-like Modular Origami Actuators	2021 – present
• Modular origami soft actuators and their transformation matrix are developed. An inverse design strategy with thickness accommodation has been proposed. • Related work has been published in Advanced Science and presented in IEEE ICRA 2023, IEEE RoboSoft 2024 and 2026.	
Artificial Muscle and Muscular-skeletal System	2023 – present
• Origami-inspired pneumatic artificial muscles with large contraction ratio enhance the motion range of the robotic arm. The muscular-skeletal system is actuated by soft origami-inspired artificial muscles, which will provide enough compliance for the robotic system, ensuring safe interaction between human and machine.	
Compact Pneumatic Controller for Large Scale Soft Actuators	2024 – present
• PneuChip with inputs can control the on/off of $m \times n$ actuators, enabling $2^{m+n} - 2^m - 2^n + 2$ distinct states. Fast and smooth state switch for the PneuChip can be realized. • Related work has been published in IEEE IROS 2025.	

PUBLICATIONS

- [1] **Z. Wang**, Y. Wang, and H. Zhang, "LEGO-like Origami Robots Standardize Structure Design of Soft Robots." *Adv. Sci.* (2025): e13881. <https://doi.org/10.1002/advs.202513881>.
- [2] **Z. Wang**, Z. Liu, Y. Wang and H. Zhang, "PneuChip: A Compact Pneumatic Controller for Large-scale Soft Artificial Muscles," *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Hangzhou, China, 2025, pp. 1425-1432, doi: 10.1109/IROS60139.2025.11246333.

- [3] **Z. Wang**, Y. Song, Z. Wang and H. Zhang, "Origami Folding Enhances Modularity and Mechanical Efficiency of Soft Actuators," *2023 IEEE International Conference on Robotics and Automation (ICRA)*, London, United Kingdom, 2023, pp. 648-654, doi: 10.1109/ICRA48891.2023.10160943.
- [4] Zhang H, **Wang Z**, Fan Z. Standardizing structure design of soft robots through origami[J]. *Nature Reviews Electrical Engineering*, 2025: 1-2.
- [5] Kako T, **Wang Z**, Mori Y, et al. 3D printable origami-inspired pneumatic soft actuator with modularized design[C]//2023 IEEE International Conference on Soft Robotics (RoboSoft). IEEE, 2023: 1-5.

PROFESSIONAL SERVICE

Reviewer

- IEEE IROS 2025
- IEEE AIM 2026